



KUBERNETES ROADMAP

Kubernetes User



by TechWorld with Nana

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Start Here



Why Learning Kubernetes is a Good Idea

Kubernetes (K8s) is being adopted at companies at a high pace and anyone with knowledge in it has a career advantage.

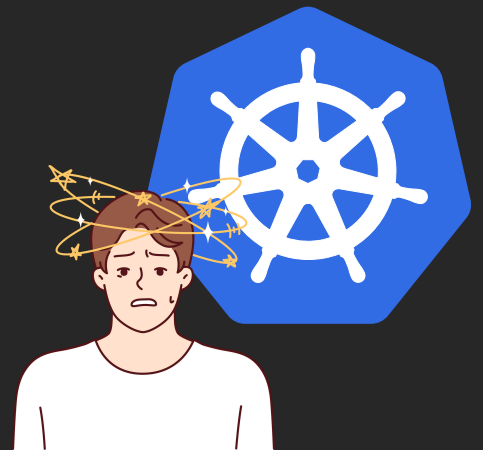
You can even become a full-time K8s engineer at your job.

Kubernetes job searches grew by 2125% in 4 years

Kubernetes is powerful, but complex

K8s is a powerful platform, but a very complex one. Which means you have a steep learning curve.

So even though many people want to learn Kubernetes, often they don't know where to start. So it certainly helps to have a guidance and sort of a roadmap of how to approach learning K8s efficiently.



That's why I want to give you this step by step roadmap of what to learn and in which order to structure your learning of K8s.

You can also find corresponding learning resources, where to learn from at the end.

Pre-Requisite - Containers

Kubernetes is built on containers, so before learning Kubernetes you should be able to run and build containers.



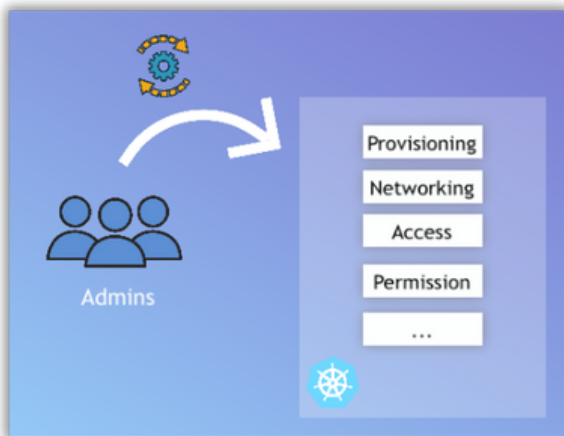
Administrator vs User

Most important point to consider, when you start learning Kubernetes

Kubernetes platform - as any other platform - has two sides or aspects:

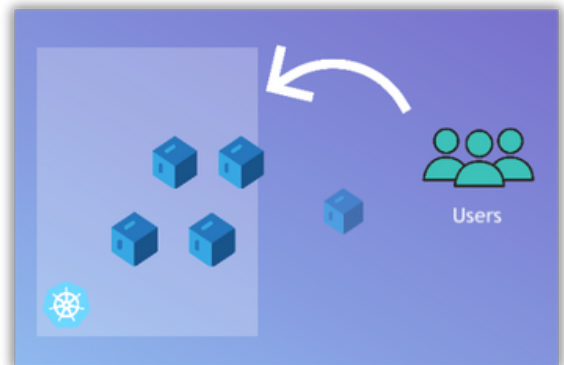
Administration -

Set Up and Maintenance of K8s cluster



Set up, initialize and configure the K8s cluster for its actual usage.

Usage of K8s cluster



Deploying applications and services into the K8s cluster

There are even different certifications for these 2:

CKA



CKAD



Administrator vs User

Decide based on your role

So these are 2 different roles and you need to understand, which aspect of the technology you want to learn?

Operations



Usage

Are you working in Operations?

You need to learn K8s Administration and Operations side



Check out the other roadmap for K8s Admin

Are you an Application Developer?

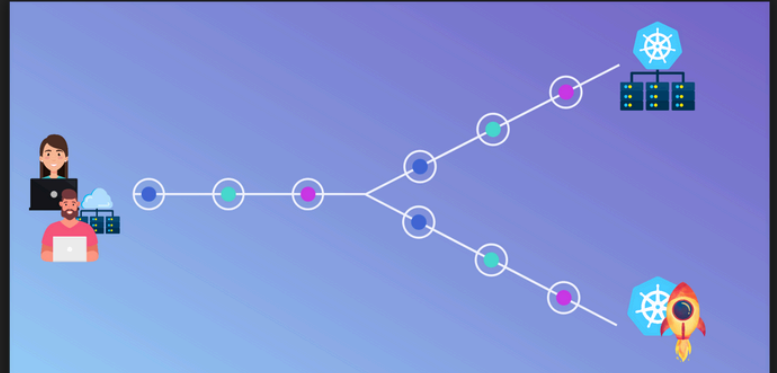
You need to learn how to use K8s to deploy and run apps inside the cluster and follow this roadmap



Administrator vs User

Learning both sides

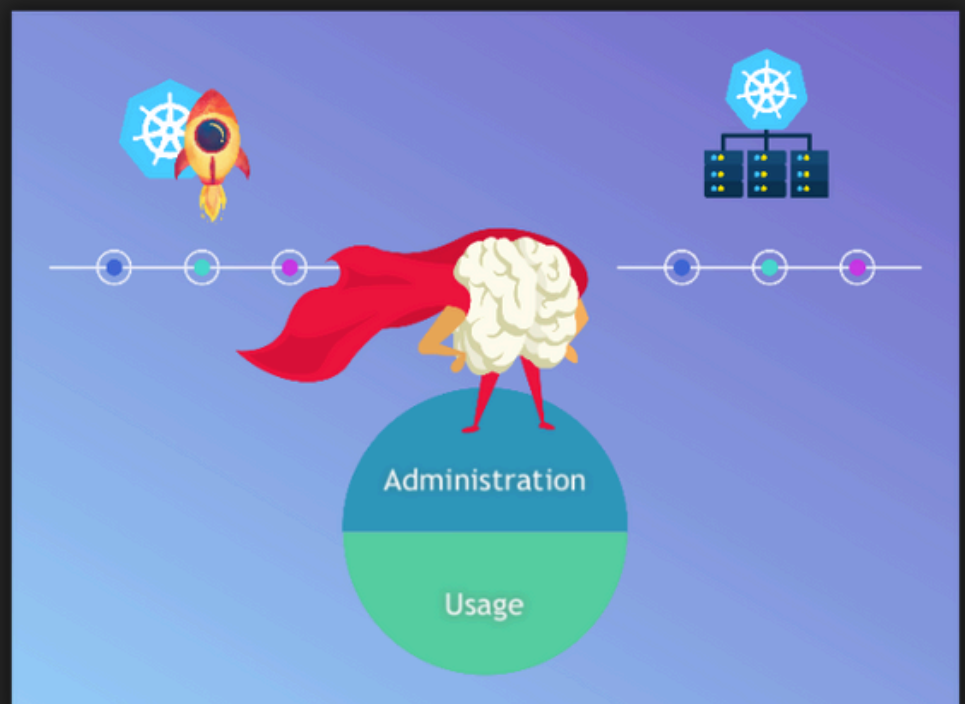
Note however, there are fundamental concepts that apply to both...



And you can eventually learn both, but you should divide it and learn one part first and then build the other on top!



This will make your learning process much easier and more structured :)



Fundamental Concepts



First, you need to learn fundamental concepts. These are the building blocks of Kubernetes knowledge, laying a solid foundation for your Kubernetes journey!

01

Learning the "Why"

why?

Most important entry point to learning any new technology is, learn the WHY behind the technology first.

- What problem does it solve, that made K8s so popular and widely used?
- The whys of how K8s helps solve issues of operating large containerized apps

02

Architecture

As part of K8s architecture, you need to learn about the **control plane and worker nodes**.

- What is the role of each?
- How are they connected?
- What processes are running on them?
- Etc



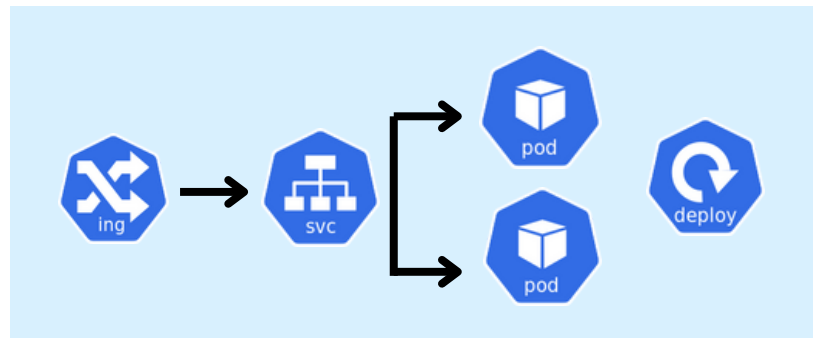
Fundamental Concepts

03

Core K8s Objects

You need to understand what are:

- Pods
- Services
- ConfigMaps
- Secrets
- Ingress
- Deployment
- StatefulSet
- Namespaces
- Volumes



These are the main K8s components. Understand why each one exists and what are their relation and connection to each other

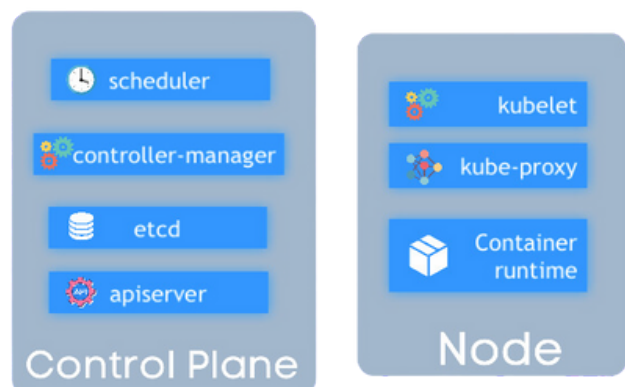
You **work with these K8s objects directly**: You create Deployments, Services etc, you join Worker Nodes to the cluster and so on.

04

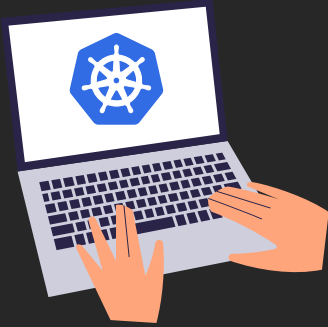
K8s Components - How K8s works behind the scenes

But there are components of K8s, that work behind the scenes and doing its magic in the cluster:

Understanding what each component does, will help you **understand the mechanisms behind the cluster**, the self-healing functionality, the way K8s manages desired vs actual state etc.



Work with Kubernetes



While understanding the concepts behind a tool is essential, after that you should get hands-on!

This gives a different level of confidence with the tool and **breaks the fear of working with it.**

05

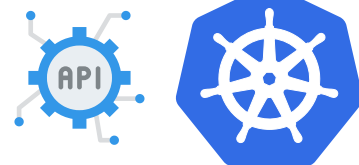
Access and Interacting with K8s

Next step is to actually access and work with a Kubernetes cluster. As part of this, you need to learn **kubectl command-line interface** and its main commands, which is essential for interacting with your Kubernetes cluster effectively.

KUBECTL



K8s cluster



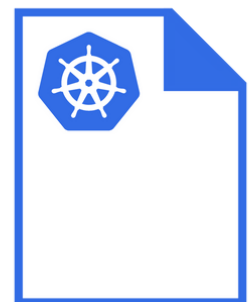
kubectl Cheat Sheet: <https://kubernetes.io/docs/reference/kubectl/cheatsheet/>

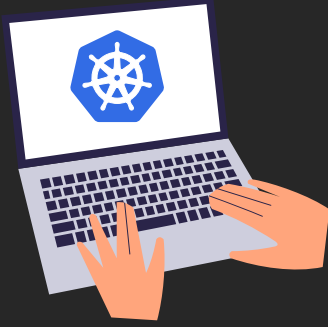
06

K8s Manifest Files

These are K8s YAML configuration files, which **describe the desired state of your applications** within the cluster. These are used to define what K8s components, like deployments, services etc, you want to create in the cluster.

Learn syntax and understand **how to create and manage these manifest files**, which will be crucial for successful Kubernetes deployments.



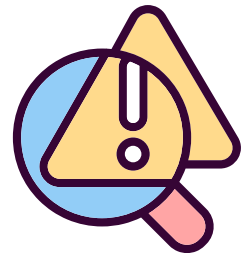


Work with Kubernetes

07

Troubleshooting

As you progress in your K8s journey and start deploying applications inside and creating various K8s resources, **you'll encounter challenges and issues** that require troubleshooting skills.



So you also need to **learn kubectl commands for troubleshooting** as well as other ways of getting insights into the cluster networking, components, applications etc:

- Check container logs
- Check cluster status
- Check Node and Pod Status
- Verify Networking
- ...



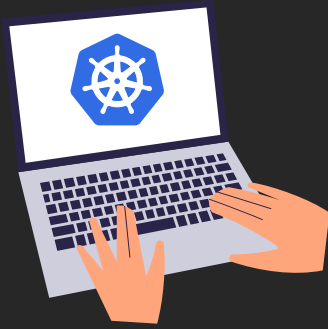
08

Common K8s misconfigurations & Bad Practices (advanced)

K8s is not opinionated and you can do a lot in the cluster, but this also means **you can do a lot wrong**, like configuring things with bad practices and misconfiguring things. That's where details of common misconfigurations will help a lot, knowing from the start what are some things that people without experience in K8s may misconfigure.

I find it easier to start with bad practices and what not to do, before learning about best practices. Because you learn exactly **what to avoid and importantly why**. This will also teach you more about how K8s works and why are things a certain way.





Work with Kubernetes

09

Helm Charts (advanced)

Another advanced topic you can learn at this point is Helm charts, especially as you start deploying third-party services in your cluster next to your own apps.

Learn:

- What is Helm
- What are Helm charts
- What are they used for etc.



Again **first why we need them and then how to use them**. This will further enhance your deployment capabilities.

On this foundation, you can now build the knowledge you need to administer and operate a k8s cluster.



Fundamentals



K8s Usage



User Roadmap

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Deep Dive on Deployment Components

As a K8s user, your main goal is to deploy and run your applications in a K8s cluster with **high availability**.

For that you need to learn:

- Concepts of replicating your app
- Scaling up/down to adjust to load
- Configure Readiness/Liveness Probes
- Achieve high availability for stateful apps as well
- Different Volume types, which one is best for your use case
- Different Service Types

Deep Dive on:

- Deployments
- ReplicaSets
- StatefulSets
- Services
- Volumes



Deep Dive on K8s Manifest Files

These K8s objects are defined in K8s manifest files, so you need deeper knowledge here as well:

- Learn different configuration options for core K8s resources
- How to define Init Containers
- How to configure different Services for your application

11

Deployment Strategies

Learn how to deploy your application changes without **causing your app downtime** or deploying without risking that the app changes will crash the entire application in production environment.

For that you need to learn different deployment strategies:

- Canary Deployment
- Blue-Green Deployment
- Rolling Updates/Rollbacks





User Roadmap

12

CI/CD Integration

As part of your developer workflow, you will have a CI/CD pipeline configured that deploys to the end environment. Instead of doing it manually.

- Learn how to integrate CI/CD with Kubernetes to do **automatic deployment**
- Learn for both: Self-Managed and Managed K8s Service



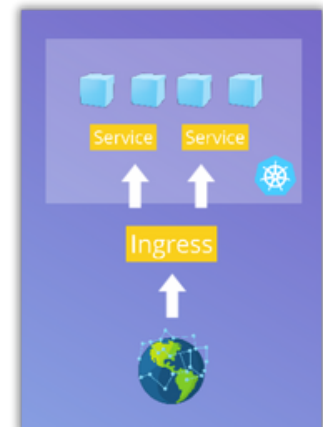
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Application Networking

Knowledge of configuration details, is even more important when deploying more complex microservices applications.

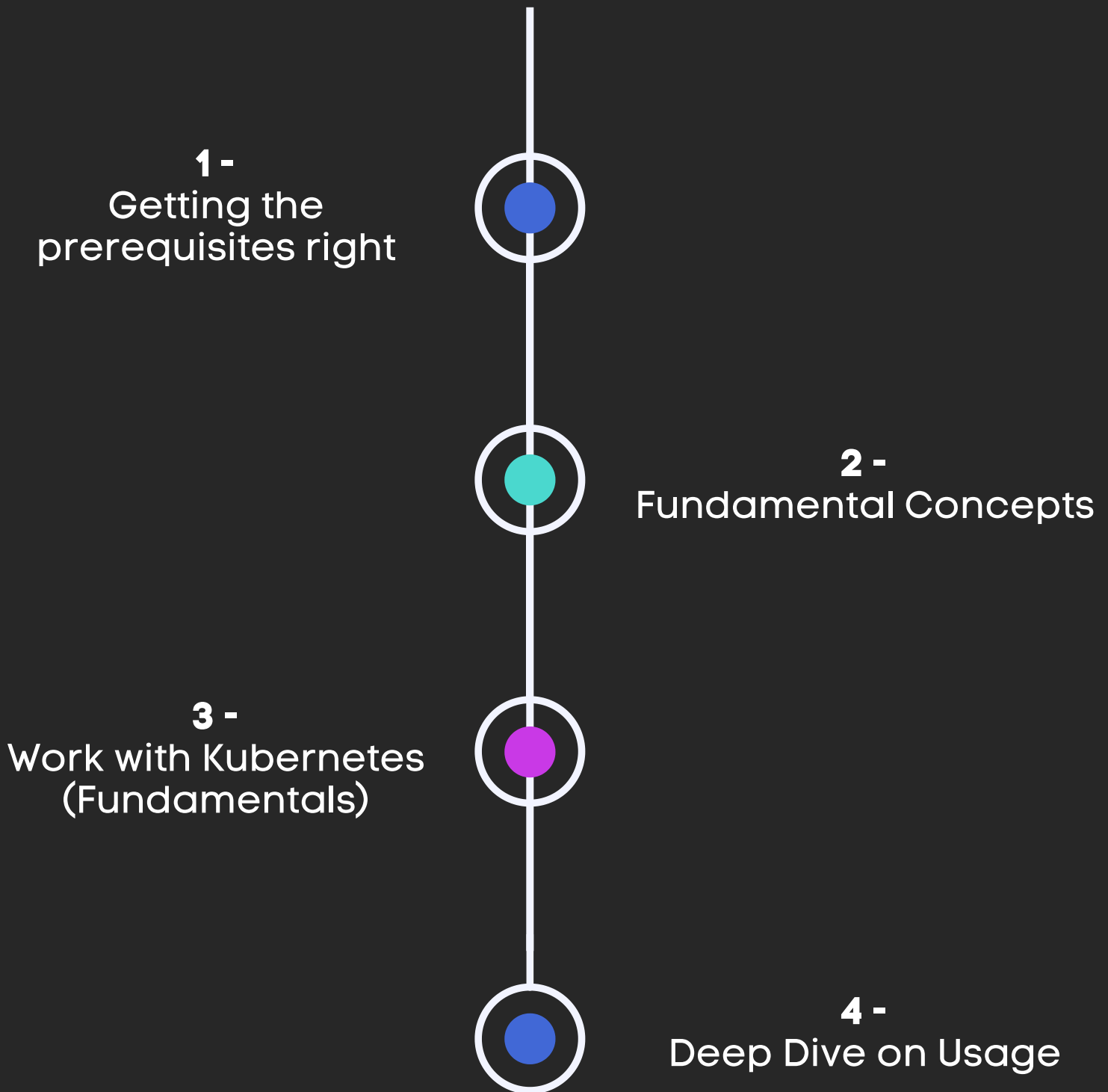
Learn how to:

- Connect them properly within the cluster and configure all needed 3rd-party services for your microservices application
- Like Service Mesh or Message Broker
- Handle traffic to the cluster (Ingress) and within the cluster



Summary

Kubernetes Roadmap

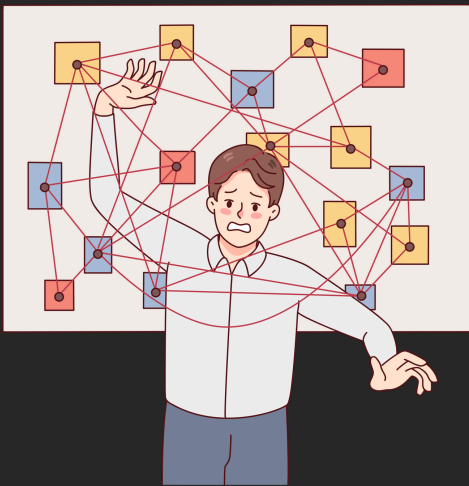


Where to learn

Official
Docs?

Blog
Articles?

Individual
Videos?



Having a roadmap like this is great for structuring your learning, but you **still need to find the right resources** that explain these concepts properly, put them in the right order, don't miss important topics etc

You can do that, but requires a lot of time and effort.

So instead, as a simpler alternative, we have created pre-structured courses, where I already did all the heavy lifting of creating the perfect curriculum, with the right hands-on projects and practical examples for this specific learning path.

So definitely check out our TWN learning resources to **make your life easier**.



TWN Learning Resources



FREE - Youtube

Kubernetes Crash Course for Absolute Beginners

You will learn everything to get started with Kubernetes, learning all the core concepts, set up K8s cluster locally and get your first hands-on experience!

Who is this course for?

For Kubernetes beginners to learn the basics



**TECHWORLD
WITH NANA**

<https://www.techworld-with-nana.com>



DevOps Bootcamp

A 6-month program to kickstart your career as a DevOps engineer. The whole bootcamp was created with the focus of making you job-ready and able to do the DevOps tasks at work

Kubernetes is a big, but just one part of it.

You will learn:

- How to USE K8s to advanced level & operational parts
- Kubernetes in combination with various other DevOps technologies
- Managed Kubernetes Service: AWS EKS
- Automating K8s cluster provisioning with Terraform
- Deploying from Jenkins CI/CD pipeline to K8s cluster
- Monitoring and Alerting in K8s
- And more

Everything based on actual real-life use cases to be able to apply it at work immediately.



<https://www.techworld-with-nana.com/devops-bootcamp>



DevSecOps Bootcamp

Integrate Security in every part of the DevOps process and advanced Kubernetes security

Security is a high priority at companies, so knowledge in this field makes you even more valuable on the job market.

It's a ~4 month program, where at the end, your acquired skills will set you apart from other engineers and make you irreplaceable for companies.



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<https://www.techworld-with-nana.com/devsecops-bootcamp>

TWN Learning Resources



Kubernetes Administrator Course

Everything you need to master the Certified Kubernetes Administrator (CKA) exam in one comprehensive and practical course!

Who is this course for?

For people:

- who want to become a Kubernetes Administrator
- who want to pass the CKA exam successfully
- who need to set up and maintain a K8s cluster at work



<https://www.techworld-with-nana.com/kubernetes-administrator-cka>

Last Learning Tip :)

Regardless of which resources you use to learn, be sure to follow this roadmap step by step

And for each concept:



1st:

Understand the WHY



2nd:

Immediately put it into practice

Good luck on your
Kubernetes journey!

